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# Georeferenced Cross-View 3D Reconstruction from Ground and Satellite Images

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## Abstract

Georeferenced 3D reconstruction from ground images requires recovering not only the local scene geometry, but also localizing every camera center and 3D point in a geographic coordinate frame. Existing cross-view geo-localization methods are able to estimate the GPS coordinate of a single ground camera, but do not reconstruct dense 3D geometry, while recent feed-forward multi-view 3D reconstruction models predict point maps and camera poses only in a local coordinate frame. We introduce X-VGGT, a feed-forward cross-view geometry transformer that unifies these tasks by jointly processing a set of ground-level images and a georeferenced satellite image to predict ground-view camera poses, dense 3D point maps, and a scene-level transform parameterized by gravity, scale, yaw and translation. This transform projects the local reconstruction into the satellite plane, resulting in the geo-registration of all ground images with the georeferenced satellite image. We further propose an evaluation protocol for this joint task and show that, across multiple cross-view datasets, X-VGGT produces accurate georeferenced reconstructions and outperforms prior cross-view geo-localization and multi-view 3D reconstruction models.

## 1 Introduction

Ground-level images are easy to collect and can be used to reconstruct detailed 3D models; however, image-based reconstruction methods typically recover geometry only up to an unknown global scale. In recent years, high-resolution satellite images have become widely available across many regions of the world. Since these images are often georeferenced, their image coordinates can be mapped to real-world geographic locations, providing GPS information for each pixel. This makes them an ideal complementary source of information for image-based 3D reconstruction, as they can provide not only a metric reference for resolving scale ambiguity, but also a global coordinate frame for localizing the reconstructed cameras and 3D points.

However, using satellite images as a geographic reference requires solving a difficult cross-view registration problem: ground-level and satellite images observe the same scene from drastically different viewpoints, with large changes in appearance, scale, and visible structure. A model that can overcome this gap would enable practical applications in autonomous driving and robotics in GPS-denied areas, disaster response from localized post-event imagery, and infrastructure inspection, where observed assets and defects could be placed directly on a map.

Previous work addresses important parts of this problem, but not the full setting. Cross-view geo-localization methods are trained to estimate ground-view camera locations within a satellite image [30, 45, 32, 31, 49, 50, 19], but they are not designed to localize individual image pixels, which is known as geo-registration, that requires additional 3D understanding. In parallel, recent feed-forward 3D reconstruction models can recover camera poses and 3D point maps from multiple views

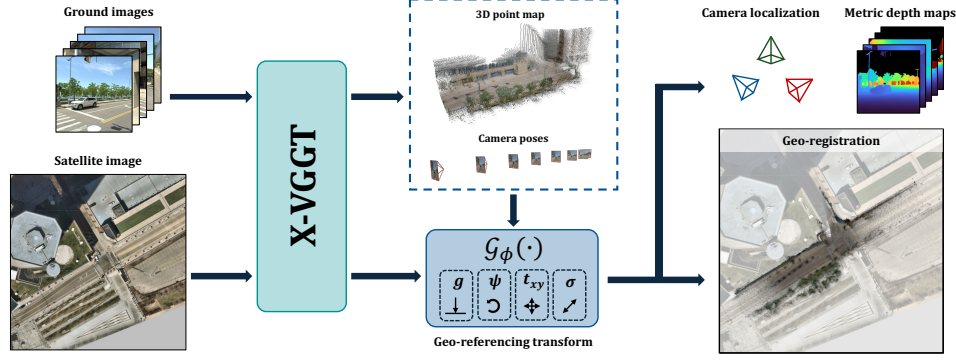


Figure 1: **Georeferenced cross-view 3D reconstruction.** Given a set of ground-level images and a georeferenced satellite view, **X-VGGT** predicts camera poses, 3D point-maps, and a georeferencing transform  $\mathcal{G}_\phi(\cdot)$ . By projecting the 3D reconstruction into the satellite plane, our model can geo-register every pixel in the ground images, localize their cameras, and generate metric depth maps.

[44, 20, 41, 22, 46], but they typically generate predictions in a local coordinate frame and are brittle under extreme viewpoint changes.

Georeferenced cross-view 3D reconstruction lies at the intersection of these two problems: using both ground and satellite images, the goal is to generate a metric 3D reconstruction where all ground-level image pixels are localized in a geographic coordinate frame. To address this problem, we introduce **X-VGGT**, a Cross(**X**)-View **G**eometry **G**rounded **T**ransformer for georeferenced cross-view 3D reconstruction. Given a set of ground images and a georeferenced satellite image, X-VGGT predicts ground-view camera extrinsics, intrinsics, metric dense 3D point maps, and a scene-level satellite alignment transform used to geo-localize image pixels and camera centers. The satellite image is processed jointly with the ground images through a multi-view transformer, allowing them to exchange information. Then, a dedicated Satellite Alignment Head predicts a ground-to-satellite coordinate transform parameterized by gravity, yaw, scale, and translation, which projects the reconstructed camera centers and dense point maps into the georeferenced satellite plane, thus enabling dense pixel-level geo-registration.

In summary, our key contributions are as follows:

- We propose X-VGGT, a feed-forward cross-view 3D reconstruction model that jointly uses ground images and a georeferenced satellite image to predict cameras, metric 3D point maps, and a satellite-ground alignment transform that geo-localizes all ground image pixels.
- We enable georeferenced 3D reconstruction from flexible sets of ground-view inputs: X-VGGT grounds camera centers and dense point maps in the satellite coordinate frame while supporting ordered sequences or unordered sets of limited-FOV images.
- We introduce a Satellite Alignment Head and georeferencing losses that directly supervise the ground-to-satellite transformation, projected camera centers, and pixel-wise geo-registration.
- We introduce new training and evaluation protocols for cross-view 3D reconstruction, using ordered and unordered sets of frames from unseen locations to evaluate the generalization of X-VGGT and competing baselines.

## 2 Related Work

**Geo-localization** aims to estimate where an image was captured by predicting its GPS coordinates from visual content alone. Existing methods are broadly grouped into classification, retrieval, and registration-based approaches. Classification methods discretize the Earth into geographic cells and use the centroid of the predicted geocell as the estimated GPS coordinate [48, 37, 27, 23, 26, 10]. Retrieval methods embed the query into a high-dimensional feature space and compare it against a georeferenced gallery, formed either by encoding GPS coordinates with a location encoder [36, 18, 28, 5, 14, 4, 29], or by using images with known locations.

Image-retrieval methods can be divided into same-view and cross-view settings. Same-view visual place recognition (VPR) embeds query and gallery images with a shared image encoder and ranks the gallery by feature similarity [34, 3, 6, 1, 2, 16, 7, 35]. Cross-view geo-localization instead matches ground-level queries against satellite images, using either shared or view-specific encoders. Prior work has improved this framework with ground-view warping, hard negative mining, re-ranking, stronger backbones, and auxiliary objectives [58, 11, 21, 57, 53]. Cross-view retrieval has also been extended to videos using multi-stage pipelines and temporal aggregation [56, 39, 25]; however, these methods typically assume ordered image sequences and are less suited to unstructured multi-view scenes.

A key limitation of retrieval-based geo-localization is that predictions remain tied to the gallery discretization: the estimated GPS coordinates represent the center pixel of a query, rather than pixel-wise GPS locations of the query image. Recent cross-view localization methods address this by registering the ground view within a georeferenced satellite image, typically through pose search, BEV transformations, dense correspondence fields, or fine-grained cross-view matching [30, 49, 45, 32, 42, 19, 31, 50, 51]. However, they remain primarily 2D pose-estimation systems: most predict a single rotation and translation within a fixed satellite area from a single ground image or panorama, without reconstructing dense 3D structure or placing multiple cameras and point maps in a shared georeferenced frame. This motivates moving beyond local 2D registration toward georeferenced multi-view 3D reconstruction.

**Multi-view 3D reconstruction** recovers camera geometry and dense scene structure rather than only a 2D pose. Recent feed-forward geometry models directly predict this structure from images: DUS3R [44] and MAS3R [20] estimate point maps and correspondences from uncalibrated image pairs, and several methods extend this paradigm to larger or more structured inputs through multi-view prediction, memory, recurrent state, relative-pose aggregation, or video geometry [9, 43, 12, 54]. In parallel, learning-based SfM and general multi-view reconstruction models are increasingly adapted to predict cameras, depth, point maps, visual appearance, or metric 3D from sparse or dense scenes [40, 41, 22, 46, 52, 55, 17].

However, these models are not designed for georeferenced ground-satellite reconstruction. They typically recover geometry in a local coordinate frame with arbitrary scale and rely on sufficient visual overlap, whereas nadir satellite and street-level images differ substantially in viewpoint, appearance, and visibility. Recent extreme-view methods partially address this gap by adapting 3D reconstruction models to drone-ground pairs [38], ground images and top-down floor plans [15], or satellite, aerial, and ground views [47]. While promising, they do not directly solve ground-satellite registration with real orthorectified satellite imagery: they either rely on drone viewpoints, use non-standard top-down inputs, or are trained with synthetic perspective-projection aerial images.

These extreme-view methods make important progress, but do not fully address the setting targeted by our method: reconstructing arbitrary ordered or unordered sets of ground views directly in the coordinate frame of a georeferenced orthorectified satellite image, without using any auxiliary modalities. In this work, our aim is to connect cross-view localization with multi-view 3D reconstruction, using satellite images to geo-register all cameras and pixels in a geographic reference frame.

### 3 Method

Given a set of ground images and a georeferenced satellite image, X-VGGT predicts ground-view camera poses, dense 3D point maps, and a satellite-to-ground transform that is used to geo-register all cameras and image pixels. In this section, we first formulate the georeferenced 3D reconstruction problem (Sec. 3.1), then describe the architecture (Sec. 3.2), training objectives (Sec. 3.4), and training strategy (Sec. 3.5).

#### 3.1 Problem Formulation

Let  $I_{\text{sat}}$  denote an orthorectified georeferenced satellite image, where image coordinates can be mapped to latitude-longitude pairs, and let  $\{I_i\}_{i=1}^N$  denote  $N$  ground-level images captured within the satellite area but with unknown geo-location. The ground inputs may be an ordered sequence, an unordered set, or perspective views extracted from a panorama. X-VGGT predicts camera extrinsics  $\hat{E}_i \in \mathbb{R}^{3 \times 4}$ , intrinsics  $\hat{K}_i \in \mathbb{R}^{3 \times 3}$ , dense 3D point maps  $\hat{X}_i \in \mathbb{R}^{H \times W \times 3}$ , and a georeferencing

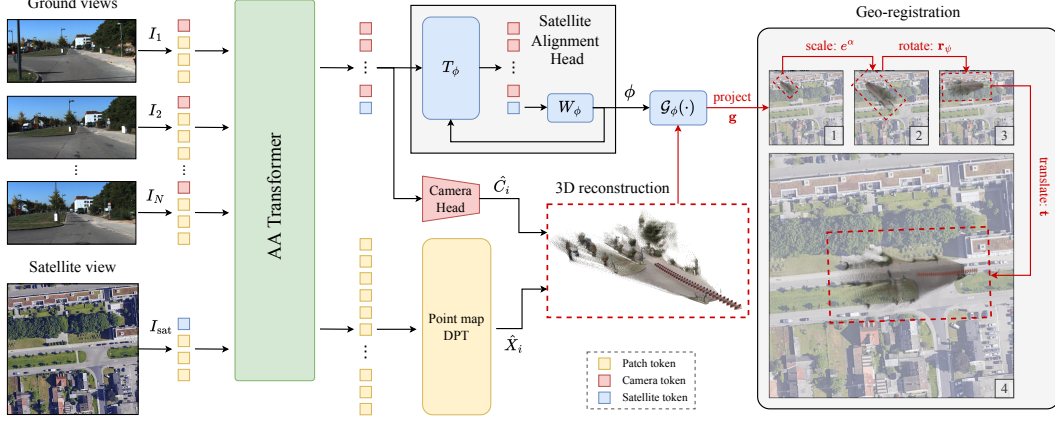


Figure 2: **X-VGGT architecture.** X-VGGT jointly processes  $N$  ground-level images and a georeferenced satellite image with an Alternating Attention (AA) Transformer. Ground-view patch tokens are decoded by the camera and DPT heads to predict camera poses  $\hat{C}_i$  and dense point maps  $\hat{X}_i$ , while the satellite token is decoded by the Satellite Alignment Head (SAH) to estimate the transformation  $\mathcal{G}_\phi$  from the local 3D reconstruction to the satellite coordinate frame.  $\mathcal{G}_\phi$  is parameterized by a gravity vector  $\mathbf{g}$  for ground-plane projection, a scalar  $\alpha$  for metric conversion, and a planar rotation  $\psi$  and translation  $\mathbf{t}$  that align the projected points with the satellite image.

transform  $\mathcal{G}_\phi : \mathbb{R}^3 \rightarrow [0, 1]^2$  used to warp pixels from the local ground-views into the satellite image plane, allowing dense point maps and camera centers to be projected and geo-localized.

The georeferencing map is a single scene-level transform. Rather than predicting each camera location independently, X-VGGT predicts one geometric map that aligns the full local reconstruction with the satellite plane. We parameterize this map by

$$\phi = (\mathbf{g}, \mathbf{r}_\psi, \alpha, \mathbf{t}), \quad \mathbf{r}_\psi = (\cos \psi, \sin \psi),$$

where  $\mathbf{g} \in \mathbb{R}^3$  is a unit gravity vector in the local reconstruction frame,  $\mathbf{r}_\psi \in \mathbb{R}^2$  is a unit yaw vector that rotates the horizontal basis,  $\alpha = \log s$  is the logarithmic scaling factor, and  $\mathbf{t} \in \mathbb{R}^2$  is a translation vector in normalized satellite coordinates.

The gravity direction defines the normal of the satellite-aligned plane. To obtain in-plane axes, we project a canonical reference direction onto the plane orthogonal to  $\mathbf{g}$ . Let  $\mathbf{a}$  be the normalized projection of the  $x$ -axis onto this plane and let  $\mathbf{b} = \mathbf{g} \times \mathbf{a}$ ; the yaw vector is used to rotate the basis as follows:

$$\mathbf{x}_\phi = \cos \psi \mathbf{a} + \sin \psi \mathbf{b}, \quad \mathbf{y}_\phi = -\sin \psi \mathbf{a} + \cos \psi \mathbf{b}.$$

For any reconstructed point  $\mathbf{X}$ , the satellite projection is thus given by

$$\mathcal{G}_\phi(\mathbf{X}) = \exp(\alpha) \begin{bmatrix} \mathbf{x}_\phi^\top \mathbf{X} \\ \mathbf{y}_\phi^\top \mathbf{X} \end{bmatrix} + \mathbf{t}.$$

This is an orthographic projection from the local reconstruction frame to the satellite plane. Coordinates are represented in normalized  $[0, 1]^2$  satellite space with a consistent upward vertical convention; metric errors are computed using the known ground sampling distance of the satellite image.

### 3.2 Cross-View Geometry Grounded Transformer

X-VGGT jointly processes the ground images and satellite image with a multi-view transformer based on alternating attention. The satellite image is appended as the final view, giving the input tuple  $(I_1, \dots, I_N, I_{\text{sat}})$  with  $M = N + 1$  total views. A shared visual encoder, initialized from DINOv2 [24], converts each image into  $P$  patch tokens  $(\mathbf{p}_i)$ . Camera ( $\mathbf{c}_i$ ) and register ( $\mathbf{r}_i$ ) tokens are appended to every view:

$$\mathbf{Z}_i^0 = [\mathbf{c}_i, \mathbf{r}_{i,1}, \dots, \mathbf{r}_{i,N_r}, \mathbf{p}_{i,1}, \dots, \mathbf{p}_{i,P}] \in \mathbb{R}^{(1+N_r+P) \times C}, \quad i = 1, \dots, M.$$

The first ground view  $I_1$ , remaining ground views  $\{I_i\}_2^N$ , and satellite view  $I_{\text{sat}}$  use separate learnable camera and register tokens, while all patch tokens are produced by the shared image encoder.

Following the alternating-attention design of VGGT [41], each block alternates between frame attention, which updates tokens within each view, and global attention, which updates tokens across the concatenated set of views:

$$\bar{Z}_i^\ell = F_\ell(Z_i^{\ell-1}), \quad Z^\ell = G_\ell([\bar{Z}_1^\ell; \dots; \bar{Z}_M^\ell]).$$

This preserves view-specific image evidence while repeatedly exchanging information across the ground and satellite views.

The satellite image has a different geometric role from a ground camera. Since the satellite image is orthorectified and treated as a nadir map, X-VGGT does not predict a perspective camera pose, finite camera center, depth map, or point map for the satellite view. Instead, the satellite tokens condition the ground 3D reconstruction through attention and are read by a dedicated Satellite Alignment Head.

**Satellite Alignment Head.** The Satellite Alignment Head (SAH) predicts the scene-level georeferencing parameters  $\phi$ . Let  $Y \in \mathbb{R}^{M \times D}$  denote the final camera-token features from all views, including the satellite token, where  $D = 2C$  because the backbone concatenates frame and global features. At each refinement step  $k$ , the head embeds the previous estimate  $\tilde{\phi}^{k-1}$ , mixes the  $M$  camera tokens with a lightweight transformer  $T_\phi$ , and reads out only the final satellite token:

$$U^k = T_\phi(Y; \tilde{\phi}^{k-1}), \quad \tilde{\phi}^k = \tilde{\phi}^{k-1} + W_\phi(U_{\text{sat}}^k), \quad k = 1, \dots, 4.$$

For the first step,  $\tilde{\phi}^0$  is represented by a learned empty georeferencing token. The final activated estimate  $\phi = A(\tilde{\phi}^4)$  normalizes  $\mathbf{g}$  and  $\mathbf{r}_\psi$  to unit length while leaving  $\alpha$  and  $\mathbf{t}$  unconstrained. Reading only from the satellite token reflects the role of  $\phi$ : it describes the alignment between the reconstructed ground scene and the satellite frame.

**3D Geometry Heads.** The ground-view geometry heads follow the VGGT design. The camera head uses an iterative transformer over the  $N$  ground camera tokens and predicts a pose encoding in  $\mathbb{R}^{N \times 9}$ , consisting of 3D translation, a 4D quaternion rotation, and 2D field-of-view parameters for each ground view. The point head is a dense prediction transformer that converts ground patch tokens into dense point maps and confidence maps,  $\hat{X} \in \mathbb{R}^{N \times H \times W \times 3}$ . Both heads receive only the ground-view tokens and the satellite view affects their predictions through the shared transformer features.

### 3.3 Geo-registration

The camera head predicts a world-to-camera transform  $\hat{E}_i = [\hat{R}_i \mid \hat{\mathbf{t}}_i]$  for each ground view in the local reconstruction frame. We recover the camera center as  $\hat{\mathbf{C}}_i = -\hat{R}_i^\top \hat{\mathbf{t}}_i$  and project both dense point maps and camera centers into the satellite image:

$$\hat{\mathbf{u}}_i(p) = \mathcal{G}_\phi(\hat{\mathbf{X}}_i(p)), \quad \hat{\mathbf{u}}_i^{\text{cam}} = \mathcal{G}_\phi(\hat{\mathbf{C}}_i),$$

where  $p$  indexes image pixels. The projected camera centers provide a fine-grained localization signal, while the projected dense points measure whether the reconstructed geometry is correctly grounded in the satellite frame.

### 3.4 Training Objectives

X-VGGT is trained with a multi-task objective that combines standard 3D reconstruction losses with new georeferencing losses. The georeferencing losses supervise the scene-level map parameters and their induced satellite projections. Let  $\phi^*$  denote the ground-truth georeferencing parameters computed in the normalized reconstruction frame. For each Satellite Alignment Head (SAH) refinement stage  $k$ , we supervise the activated estimate  $\phi^k = (\mathbf{g}^k, \mathbf{r}_\psi^k, \alpha^k, \mathbf{t}^k)$  with

$$\mathcal{L}_\phi = \frac{1}{K_\phi} \sum_{k=1}^{K_\phi} \gamma^{K_\phi - k} [1 - \langle \mathbf{g}^k, \mathbf{g}^* \rangle + 1 - \langle \mathbf{r}_\psi^k, \mathbf{r}_\psi^* \rangle + |\alpha^k - \alpha^*| + \|\mathbf{t}^k - \mathbf{t}^*\|_1],$$

where  $K_\phi = 4$  and  $\gamma = 0.6$ . The direction terms supervise the unit gravity and yaw vectors, while the remaining terms supervise scale and translation.

We also supervise the satellite projections induced by the final georeferencing transform. Specifically, we compare both the dense projected point maps and the projected camera centers against their ground-truth normalized satellite coordinates. The dense loss is computed over valid pixels  $\Omega_{\text{pts}}$ , while the camera-center loss is computed over the  $N$  ground-view cameras:

$$\mathcal{L}_{\text{dense}} = \frac{1}{|\Omega_{\text{pts}}|} \sum_{(i,p) \in \Omega_{\text{pts}}} \left\| \mathcal{G}_{\phi}(\hat{\mathbf{X}}_i(p)) - \mathbf{u}_i(p) \right\|_1, \quad \mathcal{L}_{\text{center}} = \frac{1}{N} \sum_{i=1}^N \left\| \mathcal{G}_{\phi}(\hat{\mathbf{C}}_i) - \mathbf{u}_i^{\text{cam}} \right\|_1.$$

The remaining losses follow standard feed-forward 3D reconstruction training. The camera loss  $\mathcal{L}_{\text{cam}}$  applies an L1 loss to the predicted pose encoding, decomposed into translation, quaternion rotation, and field-of-view terms. The point-map loss  $\mathcal{L}_{\text{pts}}$  combines point regression, confidence-weighted regression, and a normal consistency term over valid pixels. The final objective is given by

$$\mathcal{L} = \lambda_{\text{cam}} \mathcal{L}_{\text{cam}} + \lambda_{\text{pts}} \mathcal{L}_{\text{pts}} + \lambda_{\phi} \mathcal{L}_{\phi} + \lambda_{\text{center}} \mathcal{L}_{\text{center}} + \lambda_{\text{dense}} \mathcal{L}_{\text{dense}}.$$

All georeferencing and point-map terms have unit weight except for the camera loss, which uses a weight of 5.0.

### 3.5 Training Strategy

**Supervision signals.** Each training sample contains calibrated ground images, depth-derived point maps, ground camera poses in an East-North-Up (ENU) reference frame, and an orthorectified satellite image with known spatial resolution. During training the dataloaders project the ENU camera centers into the satellite image to obtain normalized camera-center targets  $\mathbf{u}_i^{\text{cam}} \in [0, 1]^2$ . After the standard normalization of camera poses and point maps into the local reconstruction frame, we recompute all georeferencing targets in the same coordinate system as the model predictions. Let  $E_i = [R_i \mid \mathbf{t}_i]$  be the normalized world-to-camera pose,  $\mathbf{C}_i = -R_i^{\top} \mathbf{t}_i$  the camera center, and  $\mathbf{e}_z = (0, 0, 1)^{\top}$  the ENU up direction. If  $R_0^{\text{enu}}$  is the reference-camera rotation before normalization and  $\bar{R}_0$  is the reference-camera rotation after normalization, the ground-truth gravity in the normalized reconstruction frame is  $\mathbf{g}^* = \text{norm}(\bar{R}_0^{\top} R_0^{\text{enu}} \mathbf{e}_z)$ .

Given  $\mathbf{g}^*$ , we solve for the yaw, scale, and translation that best align the normalized ground camera centers with their satellite coordinates:

$$(\mathbf{r}_{\psi}^*, \alpha^*, \mathbf{t}^*) = \arg \min_{\mathbf{r}_{\psi}, \alpha, \mathbf{t}} \sum_{i=1}^N \left\| \mathcal{G}_{(\mathbf{g}^*, \mathbf{r}_{\psi}, \alpha, \mathbf{t})}(\mathbf{C}_i) - \mathbf{u}_i^{\text{cam}} \right\|_2^2,$$

Dense satellite targets are then generated by projecting each valid depth-derived 3D point through the fitted transform  $\mathbf{u}_i(p) = \mathcal{G}_{\phi^*}(\mathbf{X}_i(p))$ , for  $p \in \Omega_{\text{pts}}$ .

**Training datasets.** We train our model with KITTI [13, 30] and a modified version of Aerial-MegaDepth [38], which we call Sat-MegaDepth. KITTI provides street-level sequences with calibrated camera poses, LiDAR-derived depth maps, and georeferenced satellite views. We construct Sat-MegaDepth from Aerial-MegaDepth [38] by adding orthorectified satellite images, which are not provided by the original dataset. Unlike KITTI, MegaDepth-style images are scattered across a scene rather than arranged as a vehicle trajectory. Training with both datasets encourages X-VGGT to handle sequential or unordered ground-view sets. Additional details on training data, satellite crops, augmentations, dataset construction, and optimization are provided in Appendix C.

## 4 Experiments

We evaluate X-VGGT on three georeferenced reconstruction tasks: camera geo-localization, pixel-wise geo-registration, and local 3D reconstruction. We first describe the datasets, scene construction, baselines, and metrics used throughout the section, then report quantitative results and ablations.

### 4.1 Evaluation Datasets

We evaluate X-VGGT on four cross-view datasets spanning different locations, sensors, and scene layouts. KITTI [13, 30] provides driving sequences with LiDAR-derived depth and georeferenced satellite images; although KITTI is used during training, evaluation is performed on the geographically

disjoint cross-area Test-2 split. Mapillary Metropolis provides synchronized multi-directional driving images, camera poses, MVS-derived depth, and satellite images, and is used for zero-shot evaluation. We further evaluate camera geo-localization on WRIVA [8] and ACC-NVS1 [33], which contain unseen locations with calibrated ground images and GPS annotations but no ground-truth depth maps. Additional dataset and satellite-image construction details are provided in Appendix D.

## 4.2 Scene Generation

We construct evaluation scenes according to the specific scene layout of each dataset: for KITTI and ACC-NVS1, we use sequences of consecutive video frames; for Mapillary Metropolis, we use consecutive multi-directional camera groups pointing along the front, back, left, and right sides of the vehicle frame; and for WRIVA, we create unordered image sets with similar camera poses. Each scene consists of  $N$  ground images and a  $150\text{m} \times 150\text{m}$  satellite view sampled around the selected ground cameras. We create satellite crops such that the centroid of the ground cameras is randomly placed within the central  $100\text{m} \times 100\text{m}$  region of the satellite image. We sample up to 1024 scenes per dataset for the main evaluation and use larger scenes with up to 96 ground images for the ablations. Additional scene construction details are provided in Appendix D.

## 4.3 Baselines

We compare against two groups of baselines. The first group consists of  $\text{FG}^2$  [50] and  $\text{Loc}^2$  [51], two cross-view geo-localization models that are trained on KITTI and have state-of-the-art performance on the cross-area split. Both estimate a 3-DoF pose in the satellite image plane, which corresponds to the camera localization and orientation.

The second group contains feed-forward 3D reconstruction models: MAS3R [20], VGGT [41], and Aerial-MegaDepth [38]. These baselines test whether generic or aerial-ground 3D reconstruction can solve the same satellite-ground alignment problem. Together, these baselines cover the two tasks X-VGGT aims to connect: cross-view geo-localization and multi-view 3D reconstruction.

Table 1: Native prediction support for each method.

| Method              | Cam. Geo. | Pixel Geo. | 3D Recon. |
|---------------------|-----------|------------|-----------|
| $\text{FG}^2$ [50]  | ✓         | ✗          | ✗         |
| $\text{Loc}^2$ [51] | ✓         | ✓          | ✗         |
| MAS3R [20]          | ✗         | ✗          | ✓         |
| VGGT [41]           | ✗         | ✗          | ✓         |
| A-MD [38]           | ✗         | ✗          | ✓         |
| <b>X-VGGT</b>       | ✓         | ✓          | ✓         |

## 4.4 Evaluation Metrics

**Camera geo-localization** measures whether the predicted camera centers and orientation are correctly grounded in the satellite image. Since X-VGGT predicts camera locations in normalized satellite coordinates  $\hat{\mathbf{u}}_i, \mathbf{u}_i \in [0, 1]^2$ , we define the location error as  $e_i^{\text{cam}} = \|(\hat{\mathbf{u}}_i - \mathbf{u}_i)\|_2 \rho_{\text{sat}}$ , where  $\rho_{\text{sat}} = 150$  is the satellite scale in meters. Orientation error is computed by projecting each camera’s principal ray into the satellite plane and measuring the angle between the predicted and ground-truth 2D directions.

**Pixel geo-registration** measures the distance between the ground truth and predicted pixel geo-locations. We compute this metric by projecting the image pixels into 3D space, and then into the satellite plane using the ground truth georeferencing transform. We report the location error between predicted and ground-truth satellite coordinates over valid pixels whose ground-truth projections lie inside the satellite image.

**3D reconstruction** evaluates the local multi-view geometry independently of satellite alignment. We align predicted camera poses to the metric ground-truth cameras with a similarity transform, then compute pairwise relative pose errors. We report relative rotation accuracy (RRA), relative translation accuracy (RTA), and their joint AUC at thresholds up to  $30^\circ$ .

## 4.5 Experimental Results

**Camera Geo-Localization.** Table 2 compares camera geo-localization across all four evaluation datasets. X-VGGT achieves the lowest camera-center and orientation errors in every setting, with average errors of 22.38 meters and  $30.93^\circ$ . Relative to the strongest competing result on each dataset, this corresponds to average improvements of 13.23 meters in localization and  $24.89^\circ$  in orientation. The gain is largest on Mapillary Metropolis, where nearby multi-directional ground views provide

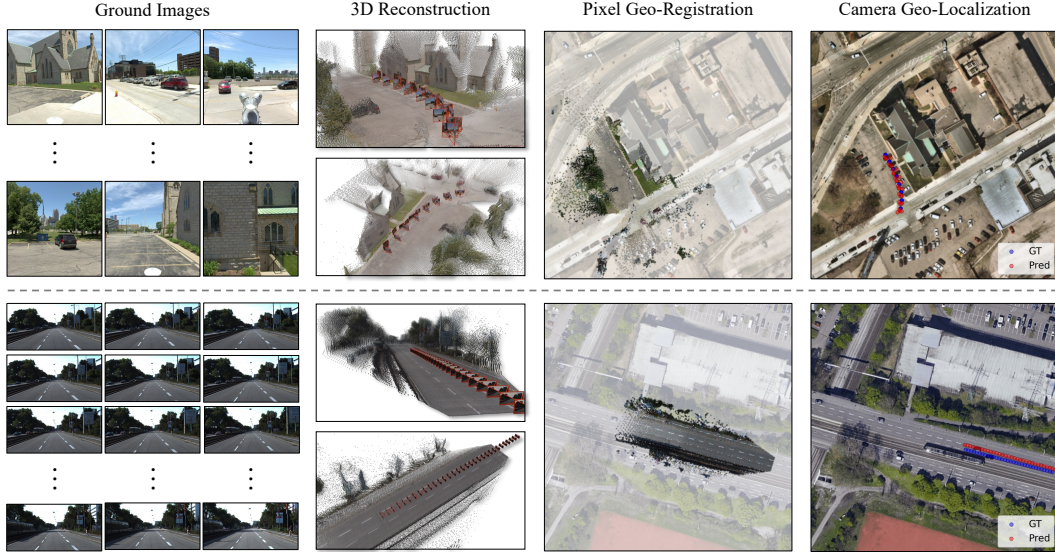


Figure 3: **Qualitative results.** We show two sample predictions from the Metropolis (top) and KITTI (bottom) datasets using X-VGGT. From left to right: input ground views, predicted 3D reconstruction, pixel-level geo-registration, and camera geo-localization. Our method jointly reconstructs the ground-view scene while projecting both image pixels and camera centers into the satellite coordinate frame.

Table 2: **Camera geo-localization.** We report mean camera-center localization error in meters and orientation error in degrees across four cross-view datasets. KITTI is evaluated on a geographically disjoint test split, while Metropolis, WRIVA, and ACC-NVS1 are fully out-of-distribution in location, sensor setup, and scene layout. X-VGGT outperforms both cross-view geo-localization and 3D reconstruction baselines on every dataset and metric, showing the benefit of jointly estimating local 3D geometry and its alignment to the satellite frame.

| Method                | KITTI        |              | Metropolis   |              | WRIVA        |              | ACC-NVS1     |              |
|-----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                       | Loc. (m)     | Orien. (°)   | Loc. (m)     | Orien. (°)   | Loc. (m)     | Orien. (°)   | Loc. (m)     | Orien. (°)   |
| FG <sup>2</sup> [50]  | 39.81        | 89.00        | 40.26        | 89.73        | 36.47        | 85.80        | 36.51        | 77.58        |
| Loc <sup>2</sup> [51] | 32.00        | 69.99        | 37.61        | 84.09        | 36.30        | 76.16        | 39.96        | 80.81        |
| MASt3R [20]           | 1103.32      | 62.22        | 1028.05      | 79.83        | 1446.36      | 88.89        | 952.52       | 86.13        |
| VGGT [41]             | 130.05       | 32.17        | 408.13       | 60.77        | 455.36       | 63.00        | 168.45       | 77.54        |
| A-MD [38]             | 104.95       | 24.68        | 154.35       | 75.82        | 412.00       | 79.24        | 190.72       | 74.84        |
| <b>X-VGGT (Ours)</b>  | <b>24.62</b> | <b>22.50</b> | <b>14.35</b> | <b>17.59</b> | <b>26.05</b> | <b>40.64</b> | <b>24.49</b> | <b>42.99</b> |

strong constraints for estimating the scene-level satellite alignment, even though it was not trained with this specific sensor layout. The cross-view localization baselines remain competitive for camera-center prediction, but produce large orientation errors. Conversely, the 3D reconstruction baselines sometimes estimate orientation more accurately, but their camera locations are unreliable because they do not predict a satellite alignment and must be projected into the satellite plane from local reconstructed poses. Overall, these results show that neither 2D cross-view localization nor local 3D reconstruction alone is sufficient; robust georeferenced reconstruction requires jointly estimating scene geometry and its alignment to the satellite frame.

**Pixel-Wise Geo-Registration.** Table 3 evaluates dense pixel geo-registration on the two datasets with depth supervision. X-VGGT achieves the lowest pixel-level localization error on both KITTI and Metropolis, improving over the strongest baseline by 18.44 meters and 71.12 meters, respectively. The closest competitor is Loc<sup>2</sup>, which benefits from a strong depth foundation model, but its dense projections remain sensitive to camera pose errors. This is most evident on Metropolis, where poor orientation estimates degrade pixel alignment, while X-VGGT remains accurate by jointly estimating geometry and satellite alignment.

Table 3: **Pixel-wise geo-registration on datasets with depth.** Mean geo-localization error (meters) over ground-image pixels for KITTI and Mapillary Metropolis.

| Method                | KITTI        | Metropolis   |
|-----------------------|--------------|--------------|
| Loc <sup>2</sup> [51] | 43.96        | 88.96        |
| MASt3R [20]           | 900.85       | 1500.67      |
| VGGT [41]             | 146.35       | 640.16       |
| A-MD [38]             | 62.07        | 319.15       |
| <b>X-VGGT (Ours)</b>  | <b>25.52</b> | <b>17.84</b> |

Table 4: **Local 3D reconstruction.** AUC of joint RRA and RTA up to 30°. X-VGGT preserves the strong local reconstruction behavior of VGGT while adding georeferencing capabilities.

| Method               | KITTI        | Metropolis   | WRIVA        | ACC-NVS1     |
|----------------------|--------------|--------------|--------------|--------------|
| MASt3R [20]          | 6.64         | 5.05         | 0.03         | 6.25         |
| VGGT [41]            | 83.29        | <b>55.31</b> | 16.90        | 67.96        |
| A-MD [38]            | 21.58        | 9.20         | 4.13         | 8.55         |
| <b>X-VGGT (Ours)</b> | <b>83.90</b> | 51.34        | <b>21.17</b> | <b>72.91</b> |

**Local 3D Reconstruction.** Table 4 shows that X-VGGT remains competitive on local 3D reconstruction, despite being trained to also estimate satellite alignment. It slightly improves over VGGT in most settings, likely because our model is initialized from the VGGT checkpoint while being further adapted to ground-level multi-view scenes. This indicates that adding georeferencing supervision does not degrade local 3D reconstruction quality.

**Qualitative Results.** Figure 3 provides qualitative examples of X-VGGT predictions. The results show that our method produces coherent local 3D reconstructions while accurately projecting both dense points and camera centers into the satellite frame.

**Ablations on the Effect of Georeferencing Losses.** Table 5 ablates the proposed georeferencing losses on the KITTI dataset. Supervising only the georeferencing parameters provides a reasonable initial alignment, but adding dense projection supervision reduces the pixel-wise geo-registration error and improves camera localization. Adding camera-center supervision further improves the alignment, yielding the lowest errors across all reported metrics. These results show that the parameter, dense, and camera-center losses provide complementary signals for reducing both camera geo-localization and pixel-wise geo-registration errors.

**Ablations on the Effect of the Number of Ground Images.** Figure 4 evaluates how performance changes with the number of input ground views on Mapillary Metropolis. Both localization and orientation errors decrease as more frames are provided, showing that X-VGGT effectively uses additional multi-view context to estimate a more accurate scene-level satellite alignment.

Table 5: **Effect of georeferencing losses on KITTI [13].** Adding dense point supervision and camera supervision improves both camera-level and pixel-level geolocation.

| Losses                          | Cam. Pose    |              | 3D Points    |
|---------------------------------|--------------|--------------|--------------|
|                                 | Loc. (m)     | Orien. (°)   | Loc. (m)     |
| $\mathcal{L}_\phi$              | 30.35        | 32.87        | 33.17        |
| + $\mathcal{L}_{\text{dense}}$  | 27.95        | 34.49        | 30.91        |
| + $\mathcal{L}_{\text{center}}$ | <b>27.24</b> | <b>30.73</b> | <b>29.45</b> |

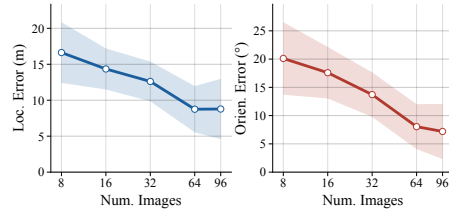


Figure 4: **Effect of the number of ground frames on Metropolis.** Additional ground views reduce both camera localization and orientation errors.

## 5 Conclusion

We introduced X-VGGT, a feed-forward model for georeferenced cross-view 3D reconstruction from ground and satellite images. X-VGGT jointly predicts ground-view cameras, dense 3D point maps, and a scene-level georeferencing transform that aligns the reconstruction with an orthorectified satellite image, enabling both camera geo-localization and dense pixel-wise geo-registration. We also proposed training and evaluation protocols covering ordered and unordered ground-view inputs across multiple datasets and sensor configurations. Experiments show that X-VGGT outperforms cross-view localization and 3D reconstruction baselines while preserving strong local 3D reconstruction quality, suggesting that robust georeferenced reconstruction requires coupling local multi-view geometry with explicit satellite-frame alignment.

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# Appendix

## A Limitations

X-VGGT assumes access to an orthorectified, georeferenced satellite image with known spatial resolution. Performance may degrade when the satellite image is outdated, low resolution, occluded, or visually inconsistent with the ground views, since the model relies on cross-view evidence to align the local reconstruction to the satellite frame.

The model is also trained with satellite crops in a specific physical scale range, roughly 150–250 meters across depending on the dataset. Applying X-VGGT to substantially smaller or larger satellite regions may change the apparent scale and context seen by the model, and would likely require fine-tuning or adapting the crop protocol.

## B Broader Impacts

This work studies georeferenced 3D reconstruction from ground-level and satellite images, unifying cross-view geo-localization with 3D reconstruction. We hope it can serve as an initial step toward treating these two tasks jointly. Potential positive impacts include improved mapping from ground images, localization in GPS-denied or GPS-degraded environments, disaster response using localized post-event images, infrastructure inspection, and robotics or autonomous navigation systems that need to connect local observations to a geographic map.

As with geo-localization methods more broadly, responsible use should consider privacy and data governance when applying the model to images of real-world places. The method should be used in accordance with dataset licenses, local regulations, and appropriate safeguards for images that may contain sensitive locations, private property, people, or vehicles. Since georeferencing errors can occur, predictions should be validated before use in safety-critical or decision-making settings.

## C Additional Training Details

### C.1 Training Satellite Images

X-VGGT is trained on two ground-satellite data sources: Sat-MegaDepth, and SatKITTI. Sat-MegaDepth extends Aerial-MegaDepth with orthorectified satellite images for each scene. Each training sample contains a set of ground images, a square satellite image, camera calibration, ground-truth 3D points, and satellite-coordinate targets for the selected ground cameras. The satellite image is resized to  $518 \times 518$  pixels before being passed to the model. Sat-MegaDepth uses crops in the 200–250m range, while SatKITTI use crops in the 150–250m range. Figure 5 shows one qualitative example our proposed Sat-MegaDepth split.

### C.2 Coordinate Targets and Augmentation

The satellite supervision uses normalized  $[0, 1]^2$  coordinates with a vertical axis convention in which larger values correspond to upward displacements in the satellite plane. Each dataloader returns normalized camera-center targets  $\mathbf{u}_i^{\text{cam}}$  and the satellite image scale in meters. During training, the satellite image is augmented with horizontal flips and random  $90^\circ$  rotations. The same transform is applied to satellite-coordinate targets, the ENU world frame, camera extrinsics, and point maps so that the georeferencing supervision remains consistent after augmentation.

Before computing the losses, we apply the VGGT-style normalization of camera extrinsics and point maps into the local reconstruction frame. Because this normalization changes the coordinate frame in which the model predicts point maps and camera poses, we recompute the ground-truth georeferencing parameters  $\phi^*$ , dense satellite targets, and valid dense projection masks after normalization. This recomputation keeps  $\mathcal{G}_{\phi^*}$ , camera-center targets, and dense point targets in the same coordinate frame as the model predictions.



Figure 5: **Example from the Sat-MegaDepth split.** The ground images at both sides are taken from the original MegaDepth dataset. We obtain high-resolution orthorectified satellite images covering the location of all ground views from Google Maps.

### C.3 Optimization

The model is initialized from the pretrained VGGT checkpoint. New satellite camera/register tokens and the Satellite Alignment Head (SAH) are optimized from the start of training to learn the cross-view alignment needed for georeferencing, while the camera and point heads retain their reconstruction roles. The top four global attention blocks are unfrozen after 10% of training, while the camera head, point head, and remaining default parameters are unfrozen after 40% of training. We use separate warmup and cosine schedules for each parameter group, with the largest learning rate for the SAH and smaller learning rates for the pretrained backbone and 3D heads. Training batches contain 4–32 ground images per scene; all images are resized to  $518 \times 518$  pixels, with zero-padding for rectangular aspect ratios. We use the AdamW optimizer with weight decay 0.05, bfloat16 automatic mixed precision, gradient clipping, and train for 80,000 iterations on four H100 GPUs.

## D Additional Evaluation Details

### D.1 Evaluation datasets.

**KITTI** [13] is an autonomous driving dataset collected around Karlsruhe, Germany, including urban roads, rural areas, and highways. The capture platform is equipped with four cameras, two grayscale and two RGB, a Velodyne LiDAR scanner, and a high-precision GPS/IMU system. We use the augmented benchmark of Shi and Li [30], which provides satellite images around the vehicle trajectory. For our evaluation, we use only the left RGB camera: the left and right stereo views are highly similar, and our goal is to evaluate sequence-level ground-satellite georeferencing rather than stereo reconstruction. We use LiDAR-derived depth maps for pixel-wise evaluation and report results on the challenging cross-area Test-2 split.

**Mapillary Metropolis** is a city-scale driving dataset collected in Chicago. It provides synchronized camera poses, georeferenced aerial imagery, panoramic street-level images, and four perspective images at each timestamp, corresponding to the front, back, left, and right directions relative to the vehicle. The dataset also includes multi-view-stereo reconstructions, MVS-derived depth maps for the perspective images, and LiDAR point clouds. Although LiDAR points are available, they are sparser than the MVS-derived depth maps for dense image evaluation, so we use the MVS depth maps when computing pixel-wise geo-registration metrics. We use Metropolis for zero-shot evaluation, since it differs from our training data in sensor layout, capture protocol, and scene distribution.

**WRIVA** is derived from the outdoor sites of the WRIVA Public Dataset [8]. Unlike KITTI and Metropolis, which are vehicle-centric driving datasets, WRIVA contains unstructured image collections from six U.S. sites with cameras scattered around each scene. The scenes have substantially different layouts from urban driving datasets, consisting largely of rural environments, open areas, and training-ground-style sites relevant to first-responder and site-modeling scenarios. WRIVA provides calibrated camera parameters and GPS annotations. Although the original dataset includes low-resolution panchromatic and multispectral satellite imagery, these modalities are outside the scope of our work, so we replace them with high-resolution orthorectified RGB satellite images from Google Maps. Since WRIVA does not provide ground-truth depth maps, we use it only for camera geo-localization and pose evaluation.

**ACC-NVS1** [33] is a multi-view dataset collected for novel view synthesis across six real-world scenes in Austin, TX and Pittsburgh, PA. It contains approximately 148K images captured from both ground and airborne viewpoints, with calibrated camera intrinsics, extrinsics, and geolocation. In our evaluation, we use the ground-level image sequences and supplement them with high-resolution orthorectified RGB satellite images to support cross-view geo-localization. Since ACC-NVS1 does not provide ground-truth depth maps, we use it only for camera geo-localization and pose evaluation.

## D.2 Evaluation scene generation.

We construct evaluation scenes using dataset-specific sampling protocols that reflect the structure of each dataset while keeping the evaluation format consistent across benchmarks.

**KITTI** scenes are generated from linear driving sequences. We sample contiguous or nearby frames from each sequence to form ordered ground-view sets, using only the left RGB camera. This produces sequence-like scenes where the model must estimate a shared satellite alignment from multiple views along the vehicle trajectory.

**Mapillary Metropolis** scenes are constructed from consecutive capture groups. Each capture group contains four perspective images aligned with the vehicle frame, corresponding to orthogonal viewing directions. We preserve this multi-directional structure when creating the scenes, which provides denser coverage around the vehicle than a single forward-facing driving sequence.

**WRIVA** contains unordered image collections rather than vehicle trajectories. We follow a VGGT-style batch construction strategy: for each scene, we choose a seed image and sample additional views with high pairwise pose similarity. This produces coherent local scenes while preserving the unordered nature of the dataset.

**ACC-NVS1** scenes are generated from ground-level image sequences. Similar to KITTI, we sample ordered frame sets from each sequence to evaluate camera geo-localization under trajectory-like motion, but on locations and capture conditions not seen during training.

**Evaluation Satellite Images.** For every evaluation scene, we compute the centroid of the selected camera centers and extract a  $150\text{m} \times 150\text{m}$  orthorectified satellite crop. Rather than placing the centroid exactly at the crop center, we randomly position it within the central  $100\text{m} \times 100\text{m}$  region. This prevents the evaluation from assuming a perfectly centered satellite crop while ensuring that the selected cameras remain inside the image. For the main results, we sample up to 1024 scenes per dataset with 16 ground images per scene. For the ablation on the number of ground images, we construct larger scenes with up to 96 ground images.

## E Additional Qualitative Examples

Figure 6 shows additional X-VGGT prediction examples across evaluation scenes. Figure 7 compares X-VGGT against baselines on camera geo-localization and shows that our model produces more accurate predictions across all datasets.

## F Licenses and Existing Assets

We use existing datasets and pretrained models, and cite the original sources throughout the paper. KITTI and Mapillary Metropolis are released under a Creative Commons Attribution-NonCommercial-ShareAlike license, and MegaDepth-derived data is released under a Creative

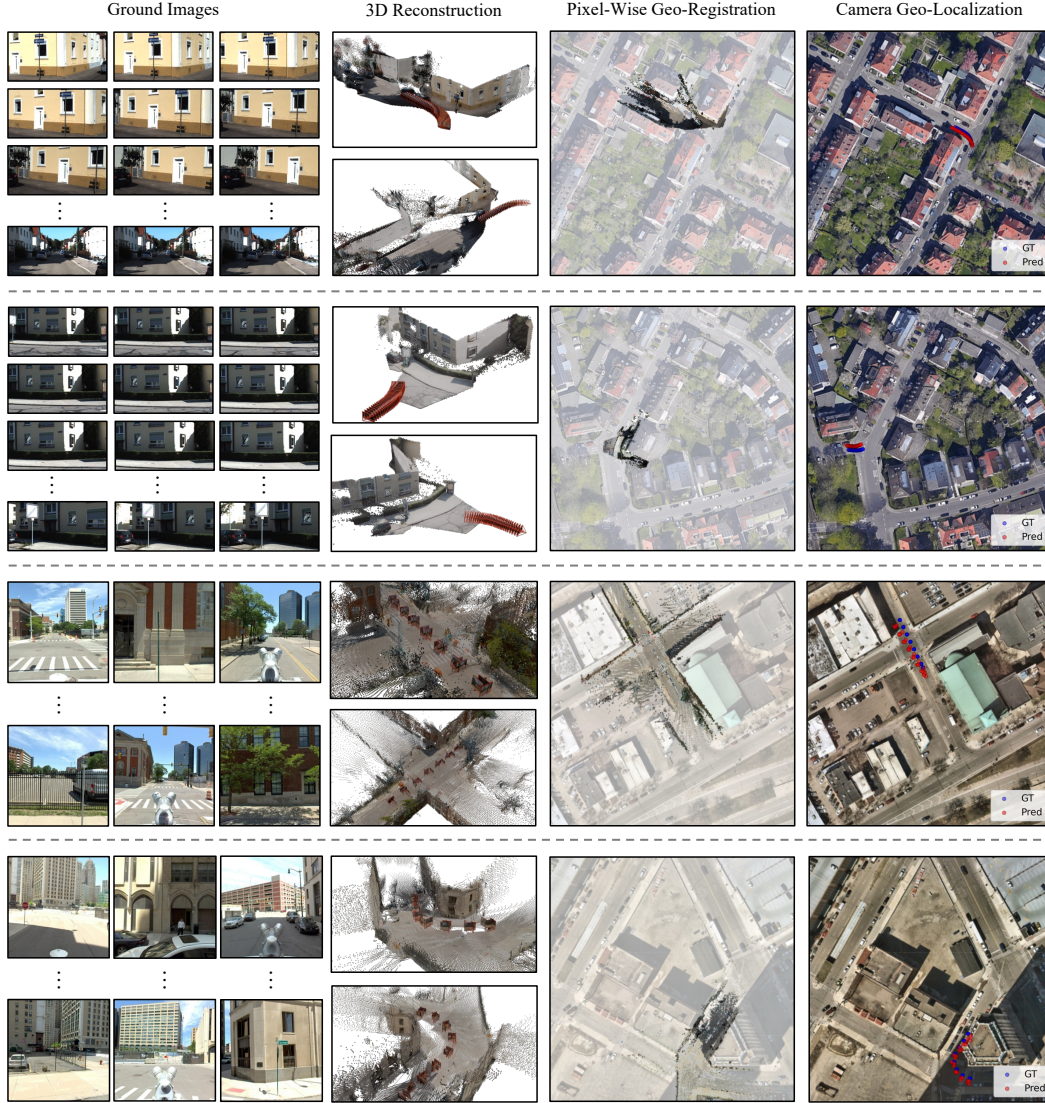


Figure 6: **Additional qualitative predictions.** We show more example predictions from X-VGGT across scenes from KITTI (first two rows) and Mapillary Metropolis (last two rows).

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Figure 7: **Qualitative comparison with baselines.** We compare X-VGGT against the baselines on camera geo-localization across the four evaluation datasets. From top to bottom: KITTI, Mapillary Metropolis, WRIVA, and ACC-NVS1.

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